Guideline for the Aerodynamic Jacobian of the Tail

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This report is a guideline for the aerodynamic Jacobian of the tail.

A. Assumptions and definitions

We assume the horizontal and vertical tails are rigid-bodies. We choose the quasi-steady aerodynamic model. Define the distance vector from the center of gravity to the aerodynamic center of the horizontal tail as $\boldsymbol{r}_{ht} = [r_{ht,x}, r_{ht,y}, r_{ht,z}]^\top$. Also, define the distance vector from the center of gravity to the aerodynamic center of the vertical tail as $\boldsymbol{r}_{vt} = [r_{vt,x}, r_{vt,y}, r_{vt,z}]^\top$. Because the aerodynamic centers of the horizontal and vertical tails are both in the aircraft symmetrical plane, then $r_{vt,y} = r_{ht,y} = 0$.

B. Local aerodynamic angles

The local velocity vector of the aerodynamic center of the horizontal tail equals

$$\boldsymbol{V}_{ht} = \boldsymbol{V}_{cg} + \boldsymbol{\omega} \times \boldsymbol{r}_{ht} \tag{1}$$

The local velocity vector of the aerodynamic center of the vertical tail equals

$$\boldsymbol{V}_{vt} = \boldsymbol{V}_{cg} + \boldsymbol{\omega} \times \boldsymbol{r}_{vt} \tag{2}$$

Using theses two equations, we can derive the nonlinear and linearized expressions for the α on the horizontal tail and the β on the vertical tail.

C. Nonlinear aerodynamic forces

Using quasi-steady strip theory, we can calculate the lift and drag caused by the vertical and horizontal tails.

Assuming small aerodynamic angles, we can project the forces into the body frame:

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$$
\begin{pmatrix} F_x \\ F_y \\ F_z \end{pmatrix} = \begin{pmatrix} -D_{ht} \\ 0 \\ -L_{ht} \end{pmatrix} + \begin{pmatrix} -D_{vt} \\ -L_{vt} \\ 0 \end{pmatrix} \tag{3}
$$

D. Aerodynamic Jacobian

In the lecture, I have shown how to derive $\frac{\partial F_z}{\partial w}$. To give you more references, let me show you how to derive $\frac{\partial F_x}{\partial v}$.

First, from Eq. (3) we know that $F_x = -D_{ht} - D_{vt}$. As you can observe from their expressions, D_{ht} is a function of V_∞, w, q, while D_{vt} is a function of V_∞, v, p, r. Therefore, we have

$$
\frac{\partial F_x}{\partial v} = -\frac{\partial D_{vt}}{\partial v} = -\frac{\partial 0.5\rho V_\infty^2 S_{vt}(C_{D0,vt} + k_{vt} C_{L_\beta}^2 \beta^2)}{\partial v} = -0.5\rho V_\infty^2 S_{vt} k_{vt} C_{L_\beta}^2 (2\beta) \frac{\partial \beta}{\partial v} \tag{4}
$$

From the lecture, we know that $\frac{\partial \beta}{\partial v} = 1/V_\infty$, thus

$$
\frac{\partial F_x}{\partial v} = -\rho V_\infty S_{vt} k_{vt} C_{L_\beta}^2 \beta \tag{5}
$$

Following the example, the partial derivatives with respective to v, w, p, q, r should be straightforward. Regarding the partial derivatives with respective to u, we assume that $\frac{\partial}{\partial u} \approx \frac{\partial}{\partial V_\infty}$. Take $\frac{\partial F_x}{\partial u}$ as an example:

$$
\begin{aligned}
\frac{\partial F_x}{\partial u} &= -\frac{\partial D_{ht}}{\partial u} - \frac{\partial D_{vt}}{\partial u} \approx -\frac{\partial D_{ht}}{\partial V_\infty} - \frac{\partial D_{vt}}{\partial V_\infty} && (6)\\
&= -\frac{\partial 0.5\rho V_\infty^2 S_{ht}(C_{D0,ht} + k_{ht} C_{L_\alpha}^2 \alpha^2)}{\partial V_\infty} - \frac{\partial 0.5\rho V_\infty^2 S_{vt}(C_{D0,vt} + k_{vt} C_{L_\beta}^2 \beta^2)}{\partial V_\infty} && (7)\\
&= -\rho V_\infty S_{ht}(C_{D0,ht} + k_{ht} C_{L_\alpha}^2 \alpha^2) - \rho V_\infty S_{vt}(C_{D0,vt} + k_{vt} C_{L_\beta}^2 \beta^2) && (8)\\
&\quad -0.5\rho V_\infty^2 S_{ht} k_{ht} C_{L_\alpha}^2 (2\alpha) \frac{\partial \alpha}{\partial V_\infty} - 0.5\rho V_\infty^2 S_{vt} k_{vt} C_{L_\beta}^2 (2\beta) \frac{\partial \beta}{\partial V_\infty} && (9)
\end{aligned}
$$

We have the terms $\frac{\partial \beta}{\partial V_\infty}$ and $\frac{\partial \alpha}{\partial V_\infty}$ because these two aerodynamic angles depends on V_∞. Using the equations that we derived in class, we have

$$
\frac{\partial \alpha}{\partial V_\infty} = -\frac{1}{V_\infty^2}(w - q \cdot r_{ht,x}), \quad \frac{\partial \beta}{\partial V_\infty} = -\frac{1}{V_\infty^2}(v - p \cdot r_{ht,z} + r \cdot r_{ht,x}) \tag{10}
$$

Therefore

$$\frac{\partial F_x}{\partial u} = -\rho V_\infty S_{ht}(C_{D0,ht} + k_{ht} C_{L_\alpha}^2 \alpha^2) - \rho V_\infty S_{vt}(C_{D0,vt} + k_{vt} C_{L_\beta}^2 \beta^2) \tag{11}$$

$$+\rho S_{ht} k_{ht} C_{L_\alpha}^2 \alpha (w - q \cdot r_{ht,x}) + \rho S_{vt} k_{vt} C_{L_\beta}^2 \beta (v - p \cdot r_{ht,z} + r \cdot r_{ht,x}) \tag{12}$$

Do not be scared by the long expression. Remember that in the steady-level flight condition $\beta_* = q_* = p_* = r_* = 0$, thus

$$\left.\frac{\partial F_x}{\partial u}\right|_* = -\rho V_\infty S_{ht}(C_{D0,ht} + k_{ht} C_{L_\alpha}^2 \alpha_*^2) - \rho V_\infty S_{vt} C_{D0,vt} + \rho S_{ht} k_{ht} C_{L_\alpha}^2 \alpha_* w_* \tag{13}$$

As another example: $F_y = -L_{\text{vt}}(V_\infty, \beta(v, p, r), \delta_r) \approx -L_{\text{vt}}(u, v, p, r, \delta_r) = -\frac{1}{2}\rho V_\infty^2 S_{\text{vt}}(C_{L_\alpha} \alpha_{\text{vt}} + C_{L_{\delta_r}} \delta_r)$, thus

$$\Delta F_y = \left.\frac{\partial F_x}{\partial u}\right|_* \Delta u + \left.\frac{\partial F_y}{\partial v}\right|_* \Delta v + \left.\frac{\partial F_y}{\partial p}\right|_* \Delta p + \left.\frac{\partial F_y}{\partial r}\right|_* \Delta r + \left.\frac{\partial F_y}{\partial \delta_r}\right|_* \delta_r \tag{14}$$

in which

$$\left.\frac{\partial F_y}{\partial p}\right|_* = -\left.\frac{\partial L_{\text{vt}}}{\partial p}\right|_* = -\left.\frac{\partial \frac{1}{2}\rho V_\infty^2 S_{\text{vt}}(C_{L_\beta}\beta_{\text{vt}} + C_{L_{\delta_r}} \delta_r)}{\partial p}\right|_* = -\frac{1}{2}\rho V_\infty^2 S_{\text{vt}} C_{L_\beta} \left.\frac{\partial \beta_{\text{vt}}}{\partial p}\right|_* \tag{15}$$

$$\left.\frac{\partial F_y}{\partial p}\right|_* = \frac{1}{2}\rho V_\infty S_{\text{vt}} C_{L_\beta} r_{\text{z,vt}}\Big|_* \tag{16}$$

Using these examples, the rest derivations should be straightforward. We will obtain the following expression:

$$\begin{pmatrix} \Delta \boldsymbol{F}_x \\ \Delta \boldsymbol{F}_y \\ \Delta \boldsymbol{F}_z \end{pmatrix}_{\text{tail}} = \begin{bmatrix} \frac{\partial F_x}{\partial u}\big|_* & \frac{\partial F_x}{\partial v}\big|_* & \frac{\partial F_x}{\partial w}\big|_* \\ \frac{\partial F_y}{\partial u}\big|_* & \frac{\partial F_y}{\partial v}\big|_* & 0 \\ \frac{\partial F_z}{\partial u}\big|_* & 0 & \frac{\partial F_z}{\partial w}\big|_* \end{bmatrix} \begin{pmatrix} \Delta u \\ \Delta v \\ \Delta w \end{pmatrix} + \begin{bmatrix} \frac{\partial F_x}{\partial p}\big|_* & \frac{\partial F_x}{\partial q}\big|_* & \frac{\partial F_x}{\partial r}\big|_* \\ \frac{\partial F_y}{\partial p}\big|_* & 0 & \frac{\partial F_y}{\partial r}\big|_* \\ 0 & \frac{\partial F_z}{\partial q}\big|_* & 0 \end{bmatrix} \begin{pmatrix} \Delta p \\ \Delta q \\ \Delta r \end{pmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \frac{\partial F_y}{\partial \delta_r}\big|_* \\ \frac{\partial F_z}{\partial \delta_e}\big|_* & 0 \end{bmatrix} \begin{pmatrix} \Delta \delta_e \\ \Delta \delta_r \end{pmatrix} \tag{17}$$

Regarding the moments, we have

$$\boldsymbol{M}_{ht} = \boldsymbol{r}_{ht} \times \boldsymbol{F}_{ht}, \quad \boldsymbol{M}_{vt} = \boldsymbol{r}_{vt} \times \boldsymbol{F}_{vt} \tag{18}$$

Then accordingly, the following expressions can be obtained:

$$\begin{pmatrix}\Delta \boldsymbol{M}_x\\ \Delta \boldsymbol{M}_y\\ \Delta \boldsymbol{M}_z\end{pmatrix}_{\text{tail}} = \begin{bmatrix}\frac{\partial M_x}{\partial u}\Big|_* & \frac{\partial M_x}{\partial v}\Big|_* & 0\\ \frac{\partial M_y}{\partial u}\Big|_* & \frac{\partial M_y}{\partial v}\Big|_* & \frac{\partial M_y}{\partial w}\Big|_*\\ \frac{\partial M_z}{\partial u}\Big|_* & \frac{\partial M_z}{\partial v}\Big|_* & 0\end{bmatrix}\begin{pmatrix}\Delta u\\ \Delta v\\ \Delta w\end{pmatrix} + \begin{bmatrix}\frac{\partial M_x}{\partial p}\Big|_* & 0 & \frac{\partial M_x}{\partial r}\Big|_*\\ \frac{\partial M_y}{\partial p}\Big|_* & \frac{\partial M_y}{\partial q}\Big|_* & \frac{\partial M_y}{\partial r}\Big|_*\\ \frac{\partial M_z}{\partial p}\Big|_* & 0 & \frac{\partial M_z}{\partial r}\Big|_*\end{bmatrix}\begin{pmatrix}\Delta p\\ \Delta q\\ \Delta r\end{pmatrix} + \begin{bmatrix}0 & \frac{\partial M_x}{\partial \delta_r}\Big|_*\\ \frac{\partial M_y}{\partial \delta_e}\Big|_* & 0\\ 0 & \frac{\partial M_z}{\partial \delta_r}\Big|_*\end{bmatrix}\begin{pmatrix}\Delta \delta_e\\ \Delta \delta_r\end{pmatrix} \tag{19}$$

Until now, the aerodynamic Jacobian of the tail has been fully derived.